Quantitative Asset Allocation Corrected Demonstrations

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Draft for author review. This note documents an in-sample maximum-Sharpe allocation and a separate rolling HRP experiment. The two experiments use different asset universes and do not establish HRP's superiority over MVO. The earlier version is retained in a labelled archive.

1. In-sample maximum-Sharpe allocation

Adjusted common-date prices for MSFT, JPM, TLT, GLD and BTC-USD run from 2021-01-04 to 2024-12-31. There are 1004 simple-return observations. Complete price rows are aligned before differencing; crypto movement over intervening weekends is included in the next common-date return.

The fitted long-only allocation maximizes estimated excess return per unit of volatility, with weights summing to one. The cash-return assumption is a constant effective annual 4%, converted consistently to a daily rate. SLSQP convergence and feasibility are checked; asset labels are preserved.

```
r[p,t] = sum[j] w[j] r[j,t]
Var[r[p]] = w' S w
Sharpe[annual] = sqrt(252) * (w' mu - rf[daily]) / sqrt(w' S w)
```

Asset	Fitted allocation	Interpretation
MSFT	22.72%	Historical in-sample numerical solution
JPM	38.53%	Historical in-sample numerical solution
TLT	0.00%	Historical in-sample numerical solution
GLD	26.93%	Historical in-sample numerical solution
BTC-USD	11.83%	Historical in-sample numerical solution

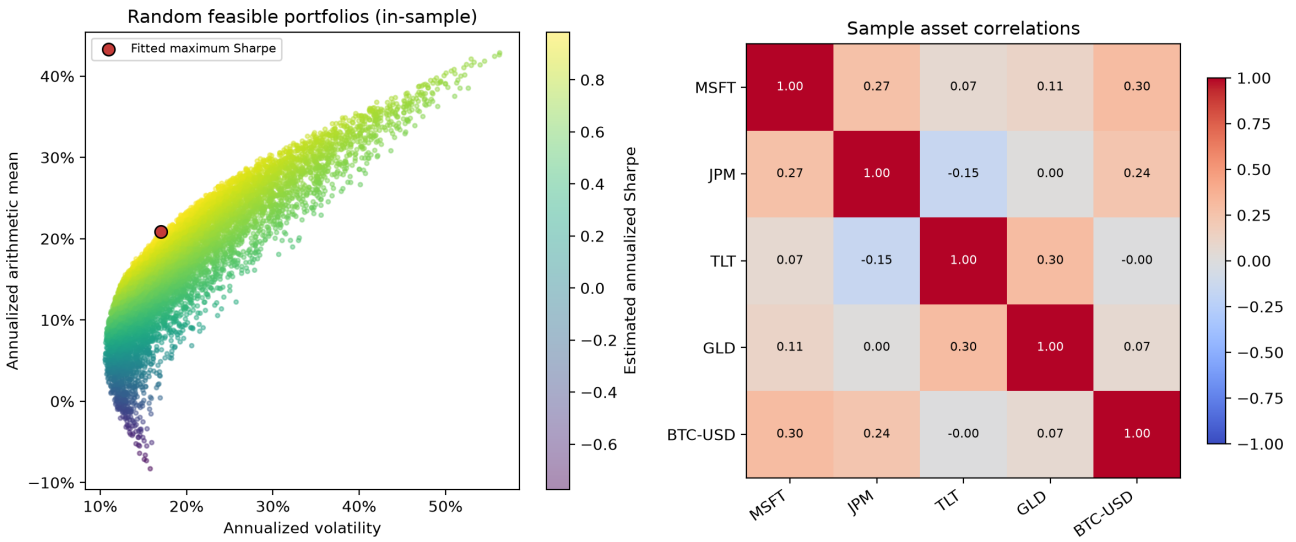


Figure 1. Seeded random feasible allocations and fitted sample correlations. The cloud evaluates weight vectors under the same estimated moments; it is neither a future-return simulation nor a computed efficient frontier.

The fitted allocation has estimated annualized arithmetic mean 20.83%, annualized volatility 17.03% and daily-excess Sharpe 0.993. Its empirical daily fifth percentile is -1.65%. These statistics use daily restored weights in the fitting sample, with no costs; they are not out-of-sample performance or a future loss guarantee.

2. Rolling HRP allocation experiment

MSFT, GOOG, JPM, XLE and GLD are the investable assets. SPY is excluded from fitting and provides a buy-and-hold market reference. Equal weighting uses the same five assets, dates, holding schedule and cost assumptions as HRP.

Each HRP target uses 252 past return observations and explicitly specified single-linkage clustering. The latest training return ends one observation before the execution close. The first test return follows that execution close. New targets are set every 21 retained observations, with holdings allowed to drift inside each block.

HRP uses correlation-based distances and covariance-based cluster variances. It avoids a full covariance inverse but still depends on historical estimates and does not impose a sector allocation cap. The implementation is supplied by PyPortfolioOpt.

Evaluation covers 2021-01-05 to 2024-12-31, with 1004 return observations. The table and chart below show the declared zero-cost case.

Strategy	Final wealth	Compound growth*	Annual vol.	Sharpe	Max. drawdown
HRP	1.594	12.41%	12.29%	0.694	-17.88%
Equal weight (same assets)	2.181	21.61%	16.01%	1.057	-20.06%
SPY buy and hold	1.683	13.95%	16.48%	0.637	-24.50%

* Compound growth uses 252/N annualization, not exact calendar elapsed time. Sharpe uses the same constant cash-return assumption. Drawdown includes initial capital.

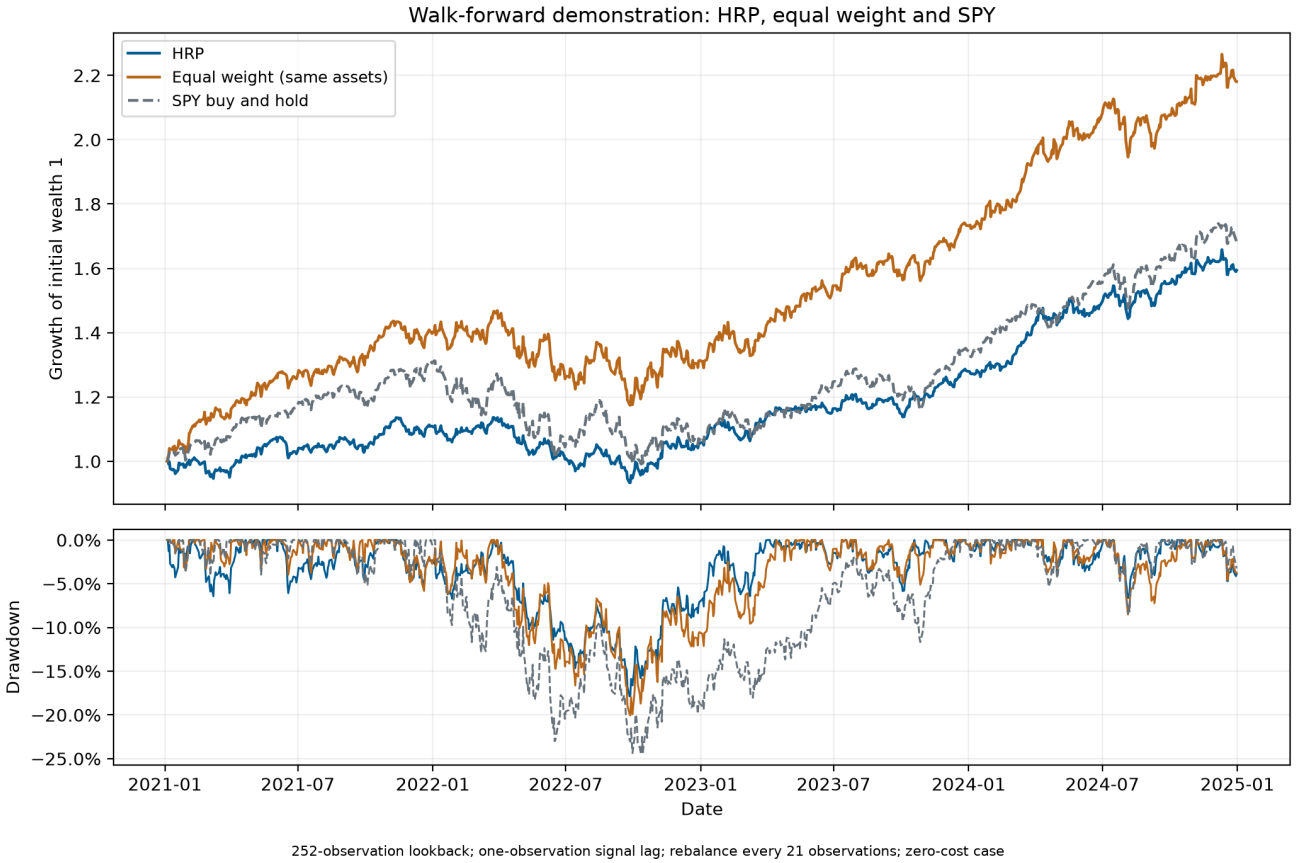


Figure 2. Out-of-window realized return accounting for the rolling rules, with initial wealth included. Equal weighting is a same-asset allocation control; SPY has a different exposure. Observed differences alone do not establish significance or general superiority.

3. Costs, corrections and limitations

A second case charges 5 basis points per dollar traded. Purchases and sales both count; initial investment from cash is included. Later trade sizes are measured against drifted prior weights. The proportional charge is applied before each new holding block. This is an illustrative sensitivity model, rather than a complete execution-cost estimate.

Strategy	Growth: 0 bps	Growth: 5 bps	Sharpe: 0 bps	Sharpe: 5 bps
HRP	12.41%	12.34%	0.694	0.689
Equal weight (same assets)	21.61%	21.56%	1.057	1.055
SPY buy and hold	13.95%	13.94%	0.637	0.637

Corrections to the earlier version

The revision fixes asset-label alignment, mixed-calendar differencing, weighted-log-return aggregation and the mismatch between fixed daily weights and a monthly holding claim. It also adds an explicit signal delay, a same-asset baseline, cost sensitivity, numerical checks, saved execution records and narrower report language. The original archive and CHANGES.md retain the correction history.

Limits of interpretation

1. The maximum-Sharpe fit is in-sample, whereas the HRP rules are estimated on rolling past windows. Different assets and evaluation procedures prevent a matched MVO-versus-HRP conclusion.

2. A small selected universe and a single historical period do not establish future performance. The original selection chronology is unknown, and no untouched holdout or statistical superiority test is claimed.

3. Adjusted daily prices approximate total returns and execution values. Cross-market timestamps, corporate-action conventions, market impact, taxes and other execution details are not fully modeled. A signal delay makes chronology clearer but does not remove every possible source of bias.

4. Mean, covariance, volatility annualization and the constant cash return are modeling choices. Serial dependence, changing regimes and estimation uncertainty can alter their interpretation. The empirical VaR is a historical quantile, not a maximum-loss guarantee.

5. Avoiding an explicit covariance inverse is not evidence of statistically accurate risk estimates. The proposed covariance-estimation and shrinkage study remains a separate, unfinished extension.

Reproducibility and sources

Prices were retrieved at 2026-10-05T08:58:25.056430+00:00. The local CSV's SHA-256 is recorded in results/data-manifest.json. The source commit is ce87a9d13131d03699dd1b0834acd945c54bbb6d. Seed 20261005, run settings, dependency versions and numerical results are saved in results/summary.json; notebooks contain executed outputs.

López de Prado (2016), Building Diversified Portfolios that Outperform Out-of-Sample.
PyPortfolioOpt: HRP implementation.
SciPy: SLSQP documentation.
yfinance: download and adjustment conventions.

Revision status: The October implementation, checks and this explanatory draft were prepared with Codex assistance. They are supplied for the original author's review, reproduction and interpretation. This note does not claim that the author independently produced the assisted revision or completed the proposed mathematical extension.